

HOW THE DUO-SKIN WORKFLOW THINKS

THE IDEA IN ONE LINE

One AI plans both characters together. You approve the concept and every part before anything expensive is made. Code and a few kits (built once) turn the approved pictures into exact Roblox files. Every step is checked, and only the part that fails is redone.

Goal: two original Roblox characters (boy + boy, girl + girl, boy + girl or girl + boy) that clearly belong together but are never clones, where every part is yours: face, hair, clothes, accessories and body. This PDF explains the loops, the design thinking behind each choice, and how the local Windows app is being built.

4

loops

3

approval gates

5

part lanes

7

checkpoints

\$5-12

AI per duo

Sources: Roblox creator-docs (official source, 26 Sep 2026), OpenAI SDK and cookbook, ComfyUI partner-node source for Tripo/Recraft prices, Tripo developer docs. Prices are estimates and should be re-checked before spending.

DESIGN PRINCIPLES

Plan before spending

Mistakes are cheapest in the plan. The plan gets the strongest model, a separate critic, and a cheap preview before anything expensive runs.

AI paints, code builds

Models make pictures. Exact geometry, Roblox templates, layout, colours and text are handled by code and by kits built once.

One thing per call

Each image call asks for one simple asset with at most 5 rules. Long prompts fail far more often than short ones.

Measure, then ask

Code measures colours, sizes, text, triangles and shapes. AI only answers yes/no questions and compares pictures twice, swapped.

You own the taste

Three approval gates. At each you can approve, reimagine, type a change, or go back. Nothing expensive runs without your OK.

Buildable by construction

The planner can only choose what the kits and libraries can make, so an approved concept never asks for the impossible.

Fix only what fails

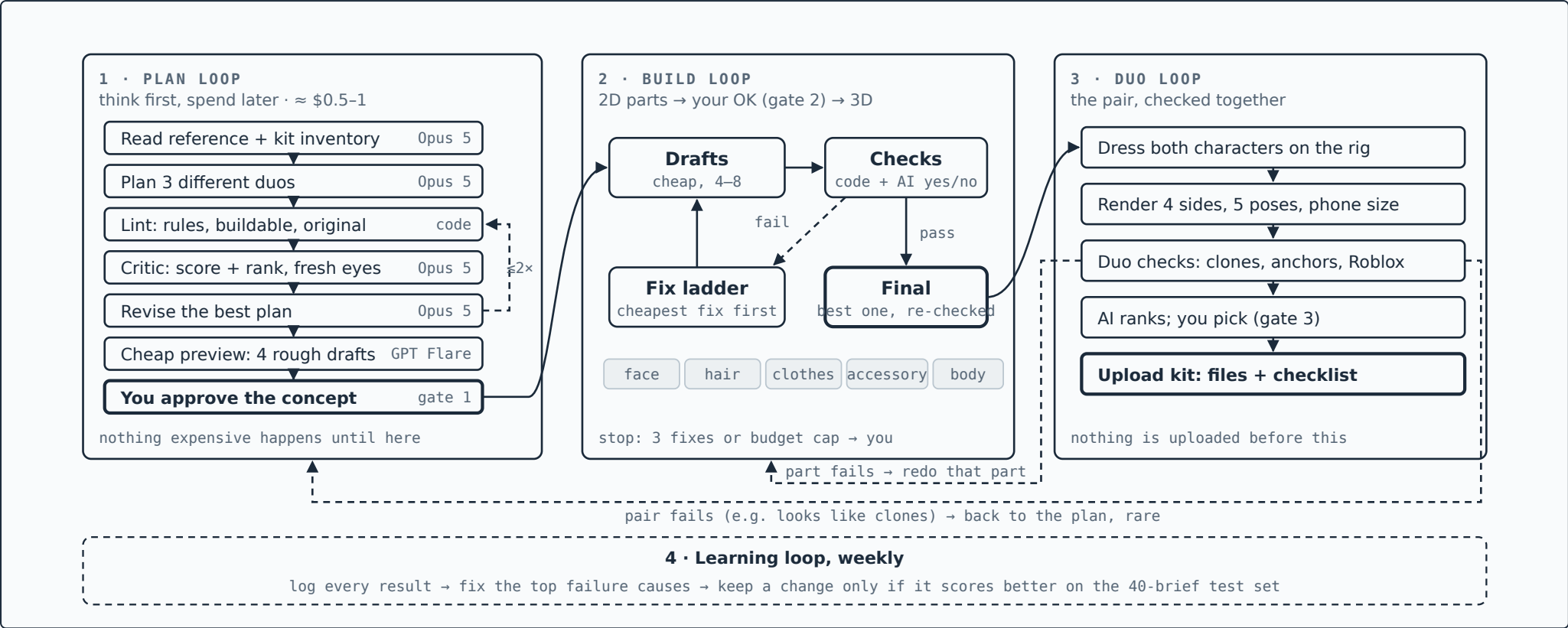
A failing part climbs a fix ladder, cheapest first, with stop rules. The rest of the duo is untouched.

Learn every week

Every result is logged. Fixes to prompts and kits are kept only if they score better on a fixed 40-brief test set.

PLAN, LOCK, BUILD, VERIFY

Four loops inside one workflow that your code controls. The plan is improved while it is still cheap text and rough drafts; after that, every part and the pair are checked, and only what fails goes back.



WHY THIS LOOP

A workflow, not one free-roaming agent

Every step is the same each time: testable, cached so you never pay twice, and resumable if something crashes. AI is used inside fixed steps with a fixed JSON format, so it can't skip a check or wander off. One autonomous agent doing everything would be slower, costlier and harder to trust.

WHY PLAN FIRST

Mistakes are cheapest in the plan

A bad idea caught in the plan costs cents. Caught after the 3D models, it costs dollars and time. So the plan gets the strongest model with extra thinking, a separate critic, and a cheap visual preview before anything is locked.

STOP RULES

No endless retries

At most 3 fixes per part and a budget cap per duo (for example \$15). If no draft passes, the technique changes rather than simply retrying. The best version so far is always kept, and anything still failing goes to you with a short report.

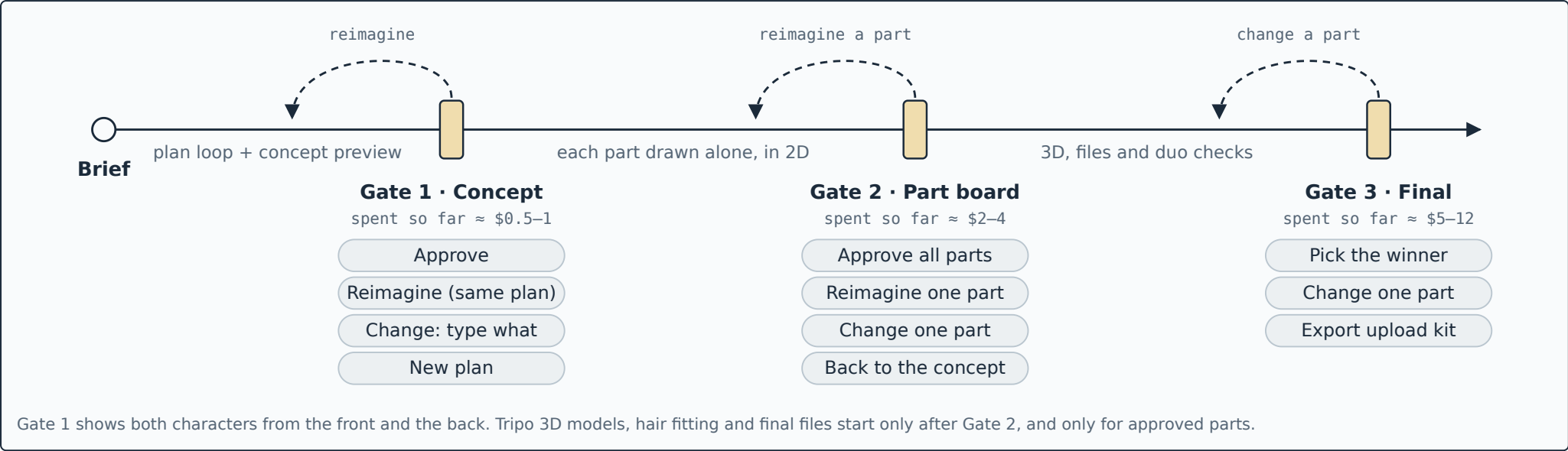
WHAT STAYS HUMAN

About 1-2 hours per duo

Approving the concept and the part board (a minute each, or reimagine), polishing hair (30-60 min until code-built hair matures), picking the winner, and uploading in Studio (Roblox has no upload API).

THREE APPROVAL GATES

Nothing expensive starts until you approve it. Each gate shows the money spent so far. Reimagine keeps the plan and makes a fresh version; Change applies only what you type.



HOW EVERY PART IS MADE

Each part uses the method that makes it look best and pass Roblox's rules. AI does the creative painting; code and kits handle exact geometry.

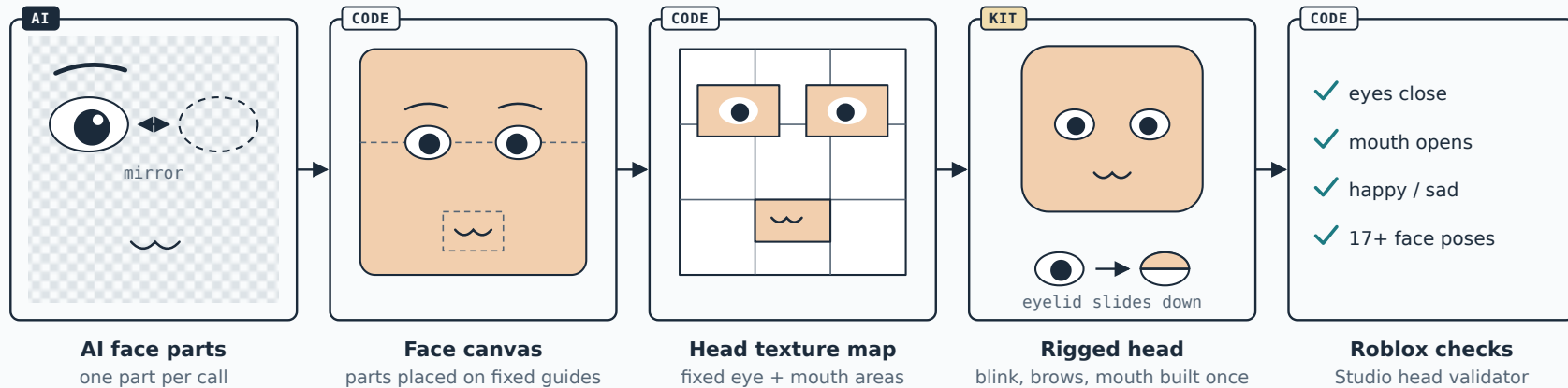
PART	HOW IT'S MADE	MAIN TOOLS	TRIP0 P2?	AI COST
Shirt + pants	AI paints prints and patches; code builds the 585 × 559 files in layers (fabric, folds, prints, painted shoes and bracelets)	GPT Image 2.5, Recraft V4.1, your code	No, these are 2D images	≈ \$0.5
Face	AI makes face parts one at a time; code paints them onto a head base that already blinks and talks	Recraft V4.1 or GPT Image 2.5, head base, your code	No, a face is paint on a moving head	≈ \$0.3–0.6
Hair	AI draws the style; AI picks the closest kit hairstyle and pieces; code fits and recolours; a person polishes	GPT Image 2.5, Tripo multiview, hair kit, Blender	Only for the 4 views; P2 as a backup for rounded styles	≈ \$0.2
Signature accessory	AI draws it from the front; Tripo makes 4 views and the 3D model; code repairs and checks it	GPT Image 2.5, Tripo P2, Blender	Yes, the main tool	≈ \$1–4
Sticker accessory	AI draws a flat sticker; code cuts its outline and turns it into a thin 3D slab	GPT Image 2.5, your code	No, code is exact	≈ \$0.1
Brows + lashes	Painted on the head as part of the face	same as the face	No	—
Body	Built once by code in your own proportions; recoloured per character with a plain modesty layer	Blender, your code	No, boxes need to be exact	≈ \$0

TRIP0 P2 REPORT CARD

GREAT FIT Accessories Plush pets, bags, props, small hats. Main 3D tool, with a triangle target and textures.	OK AS A BASE Rounded hair Buns, bobs and fluffy shapes. Needs repair and polish, so it's the hair backup.
WEAK Spiky anime hair Tripo's own blog says thin strands fuse into blobs. Use the hair kit instead.	NOT USED Face Tripo sculpts a 3D face that can't blink. Faces are paint on the head base.
NOT USED Body A blocky body is boxes. Code makes them exact and clean.	NOT USED Clothes Shirt and pants are 2D images, so there is nothing to model in 3D.

ONE RIGGED HEAD, ENDLESS FACES

Roblox only accepts heads whose eyes close and mouth opens. So one blocky head is built once with moving eyelids, brows and mouth areas; each new face is only new paint on those areas.



BUILT ONCE · THE HEAD BASE

A head that already blinks and talks

1. You model a blocky head in your own shape, with large texture areas for the eyes, brows and mouth so lines stay sharp.
2. Small eye, brow and mouth pieces sit slightly inset in the head. Each eye has a skin-coloured lid that can slide down.
3. Roblox Studio's Avatar Setup generates the face poses and cage. A quick Blender clean-up makes blinking slide the lid instead of squashing the eye.
4. Make 2-3 eye shapes (round, narrow, sleepy) and 3-4 mouths (smile, cat mouth, open grin, smirk).
5. Studio's head check must pass: eyes close, brows raise, mouth opens, happy and sad. It's free, because Studio validates before you pay.

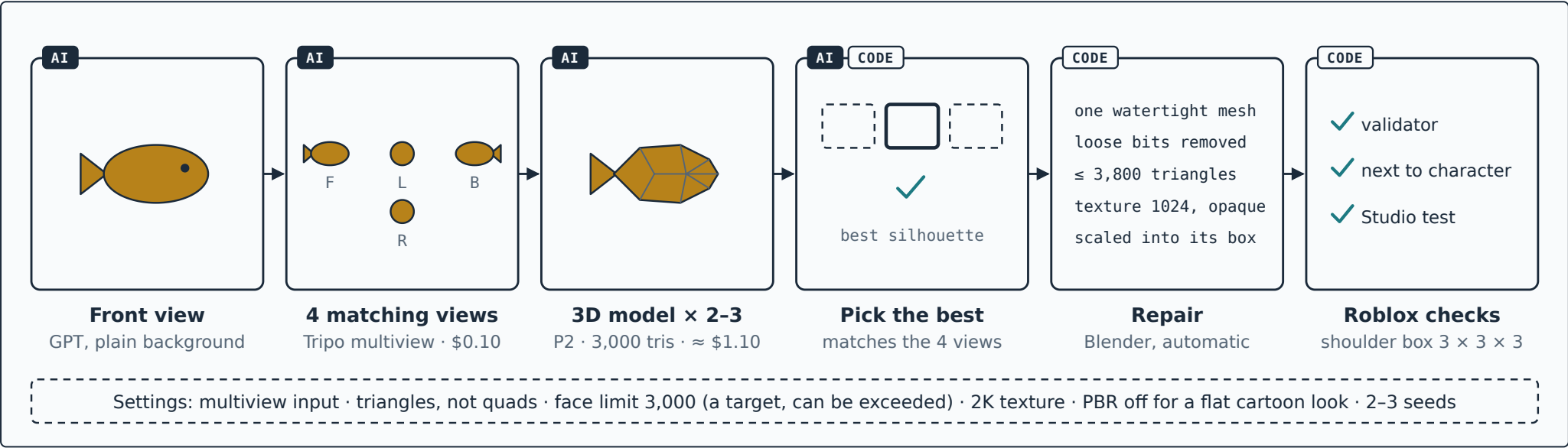
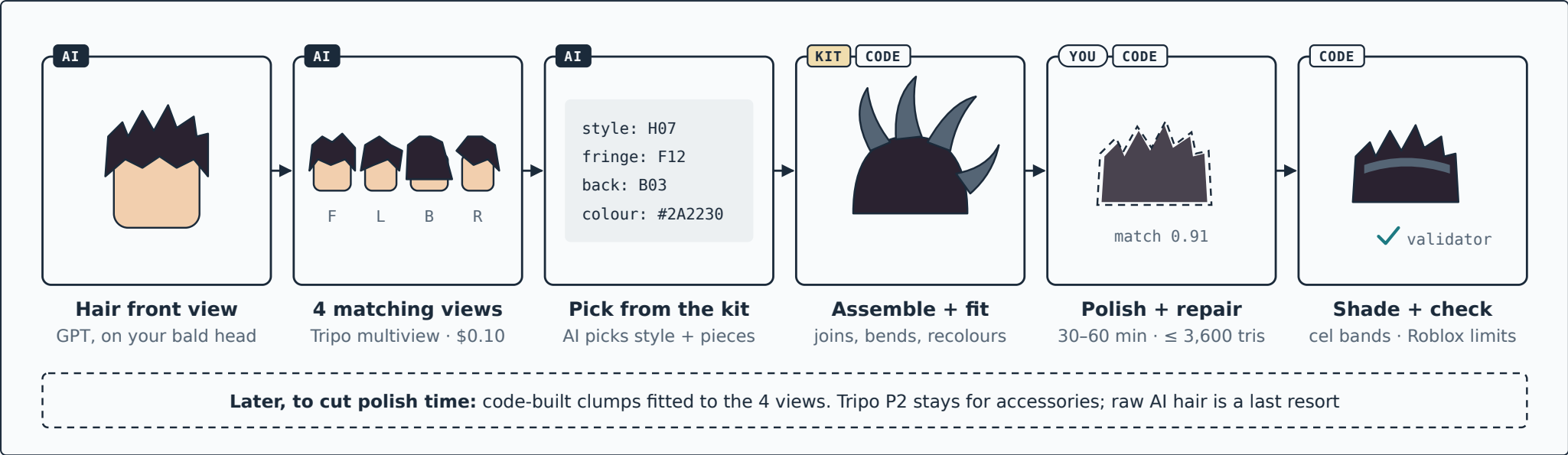
EVERY CHARACTER · AUTOMATED

A new face in minutes

1. The plan picks face features from a fixed list: eye shape, iris style, highlights, lashes, brows, mouth, cheek marks and expression. A and B must differ in at least 3.
2. AI makes each part separately (one eye, mirrored by code; brows; mouth; the closed-eye line shown when the lid comes down; the inside of the open mouth).
3. Code places the parts on fixed guides, snaps colours to the palette and evens out line thickness.
4. Code wraps the face onto the head's texture map using a lookup table made once.
5. Code renders 5 poses (neutral, blink, mouth open, happy, sad) at full size and phone size. An AI checks for smearing, uncovered eyes and symmetry.
6. The face registry blocks any face too close to one made before. You pick from the top 2-3.

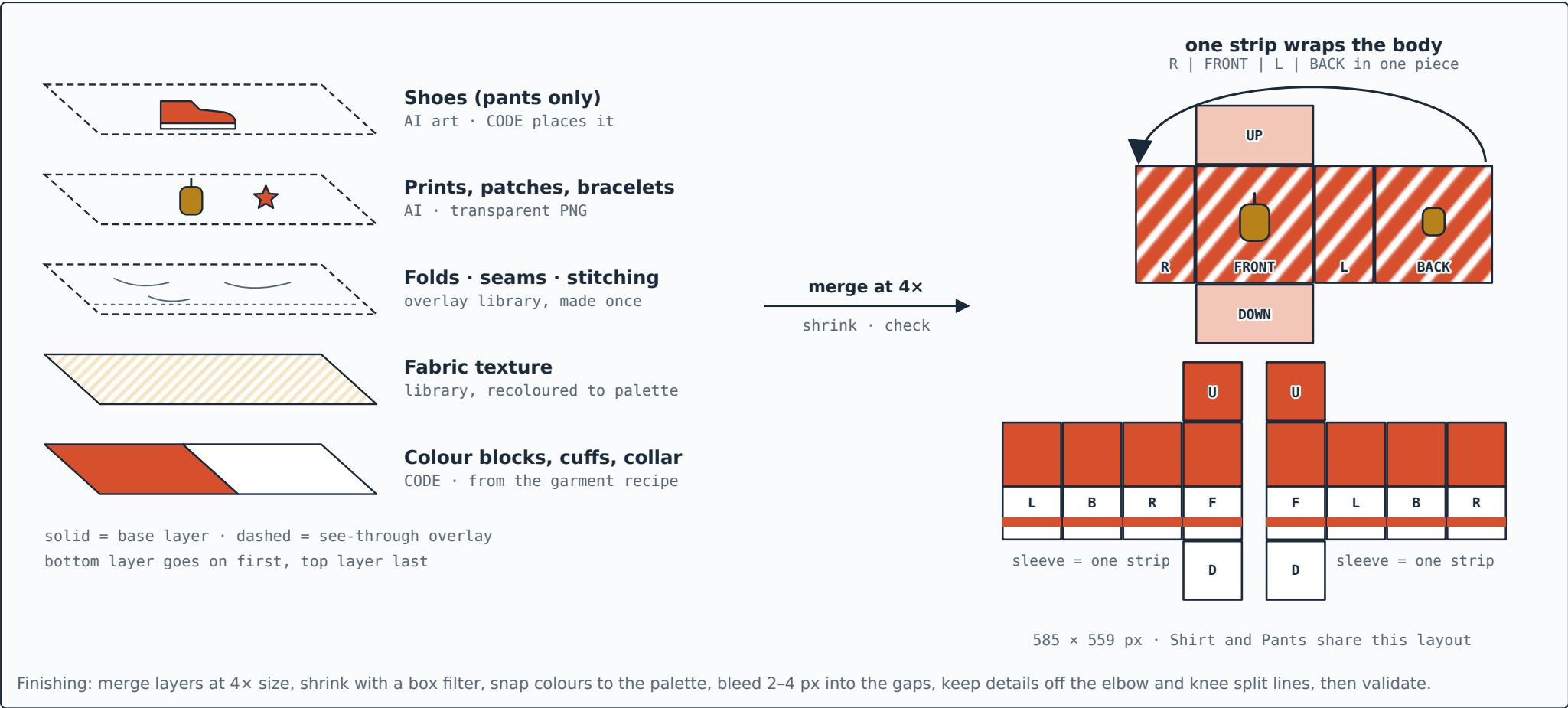
HAIR FROM HUMAN-MADE STYLES; ACCESSORIES FROM TRIPO P2

AI 3D tools melt thin anime spikes into blobs, so hair starts from artist-made styles that AI picks and code fits. Accessories are Tripo P2's strength: 4 matching views in, a clean small model out, then automatic repair and Roblox checks.



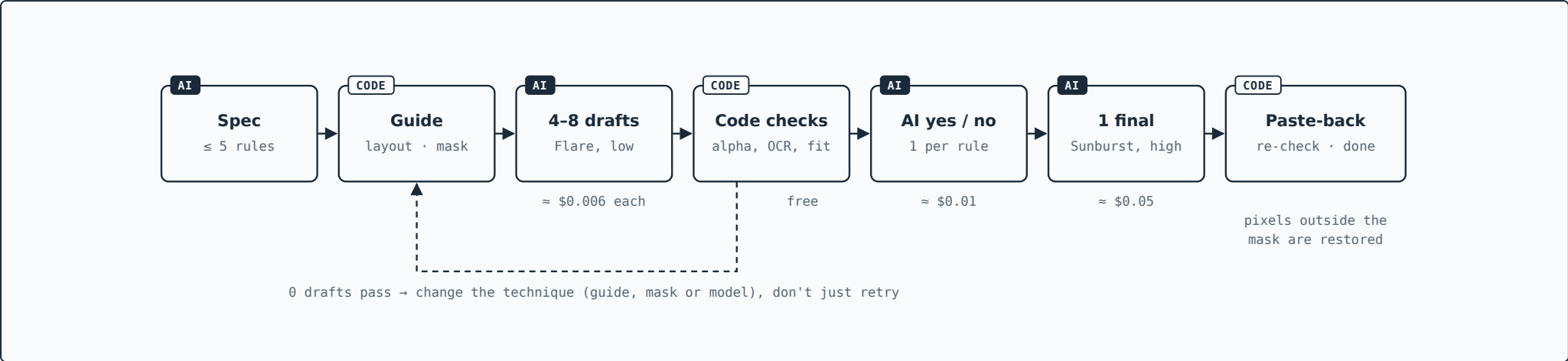
LAYERS IN, EXACT TEMPLATE OUT

No image model paints the 585 × 559 Roblox template correctly, so the AI makes prints, patches and shoe art, and code places everything into the exact template regions (checked against Roblox's official files).



THE NO-WASTE IMAGE LOOP AND PROMPT RULES

Image models get single rules right most of the time but whole long prompts right only about a third of the time. So each call is simple, drafts are cheap, checks are automatic, and only the best draft is paid for at high quality.



UNIVERSAL PROMPT RULES

1. One asset per call, no more than 5 rules in a prompt.
2. Code draws the layout (guide canvas, masks, silhouettes); the model only paints inside it.
3. Reference images are named by position ("Image 1 is the style reference, Image 2 is the guide").
4. Say what you want, not what you do not want; forbidden things are listed once, briefly.
5. Never ask for text in an image; any text is added by code, and OCR rejects stray text.
6. Edits carry a "keep everything else exactly the same" list, and code restores pixels outside the mask.
7. Transparent assets never describe a background; code checks the alpha channel for real transparency.
8. The same 1–3 style references go into every call, so the style does not drift.
9. Side and back views come from a 3D-aware tool (Tripo 4-view), not from a general image model.
10. Exact model versions are pinned; a model upgrade must pass the test set first.

ASSET	BEST TOOL	HOW IT STAYS ON TARGET
Duo concept	GPT Image 2.5 Sunburst, edit mode	repaints a code-drawn picture of your two bodies; outlines locked, no text
Prints, patches	GPT Flare, transparent · A/B test with Recraft V4.1 vector	one graphic per call, ≤ 6 colours, no text; colours snapped by code
Face parts	Recraft V4.1 vector · A/B test with GPT mask edit	one part per call; one eye made, then mirrored by code
Hair + accessory views	GPT front view, then Tripo multiview	only the front comes from GPT; the other views come from a 3D-aware tool
Fabric	GPT Flare, then code makes it tile	flat swatch with no folds; seam fixed by shift + inpaint

Hair · in bundle
kit style + polish

Face · in bundle
AI parts on a rigged head

Shirt · own item
AI prints · bracelets painted

Pants · own item
shoes painted on

A · lead
girl · hyper energy

linked pair
lantern (waist) ↔ koi (shoulder)

B
boy · deadpan energy

Hair
different shape

Face
different features

Jacket
different cut + fabric

Cargo + shoes
different material

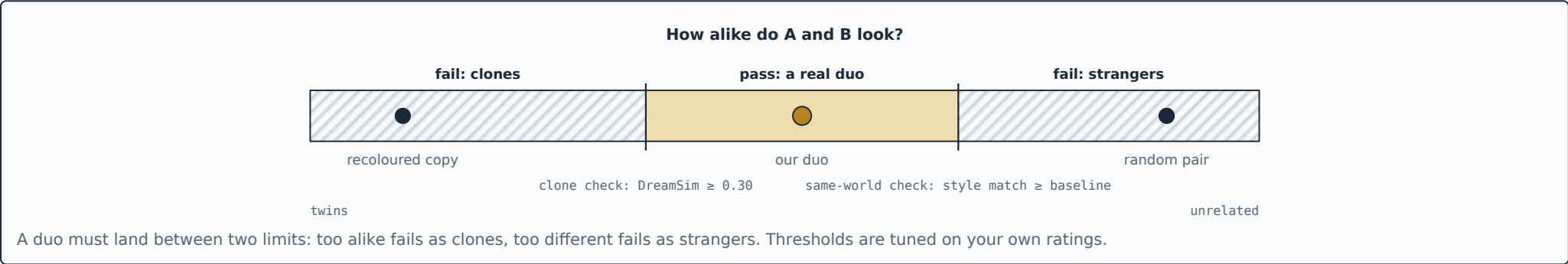
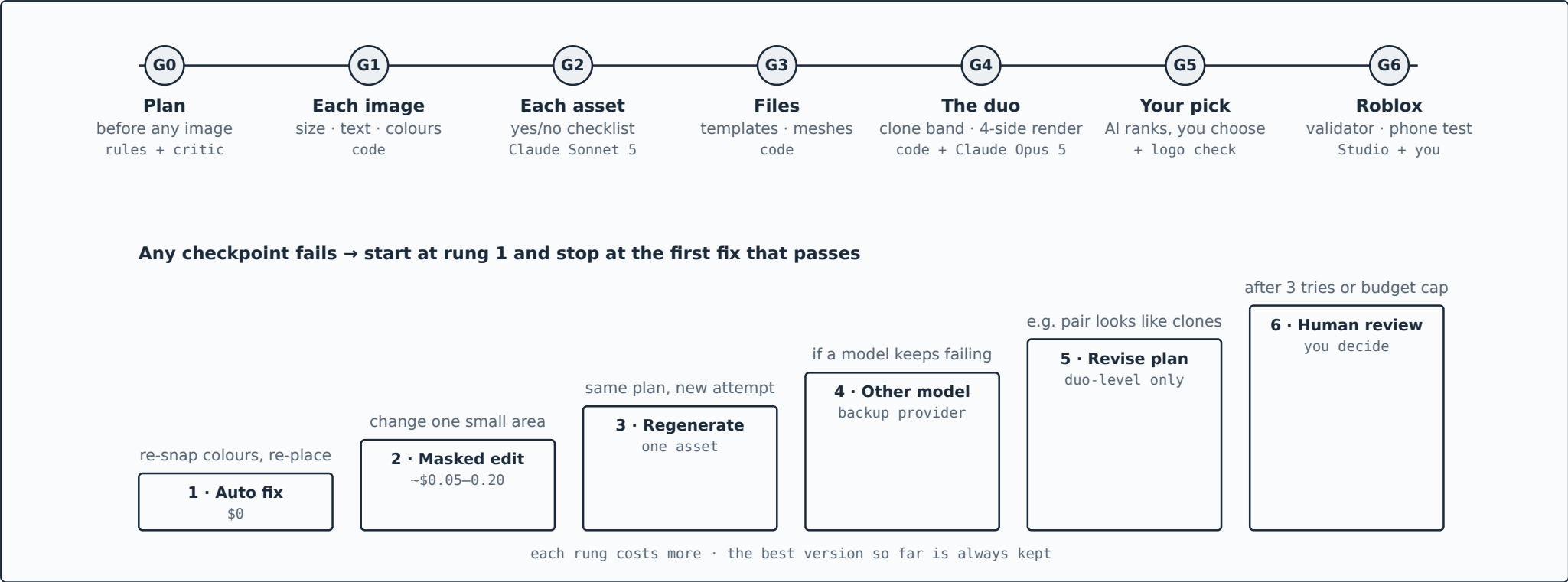
Same (2-3 anchors): lantern-gold accents · night-market theme · canvas fabric

Different (5+ axes): main colour · hair · face · outfit cut · accessory slot · energy

WHAT DIFFERS	A · LEAD (GIRL)	B (BOY)	HOW IT IS CHECKED
Main colour	 tomato red	 teal	colour distance $\Delta E \geq 15$
Hair	twin buns (kit style H03)	messy spiky crop (kit style H07)	different kit style and pieces
Face	round sparkle eyes, cat mouth	half-lidded eyes, smirk	at least 3 of 7 face features differ, and the face has never been used before
Outfit	cropped tee + pleated skirt	work jacket + cargos	outline overlap $\leq 85\%$
Main print	lantern doodles	koi on the back	print registry: never reused
Accessory	lantern keychain · waist	koi plush · shoulder	different slot, category overlap $\leq 1/3$
Energy	hyper	deadpan	plan critic

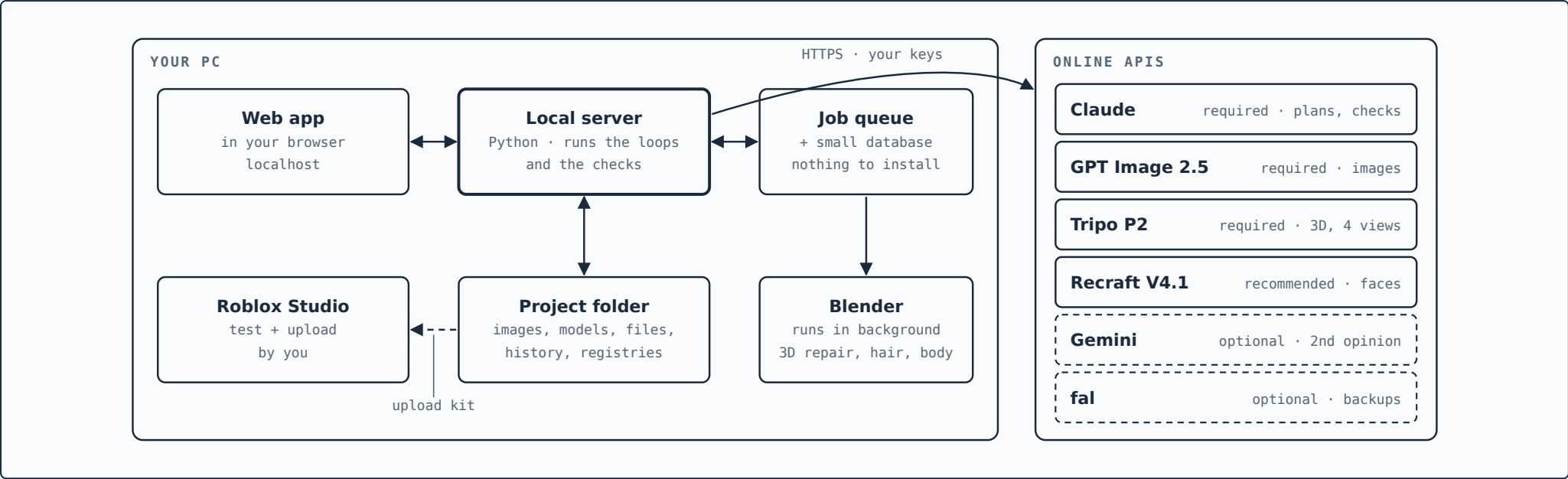
SEVEN CHECKPOINTS AND A FIX LADDER

Code measures the numbers. AI judges answer yes/no questions and compare twice with the order swapped. At the end, AI ranks and you pick. A logo and known-character check always runs; "not too close to my reference" runs only when you switch it on.



A LOCAL WEB APP ON WINDOWS

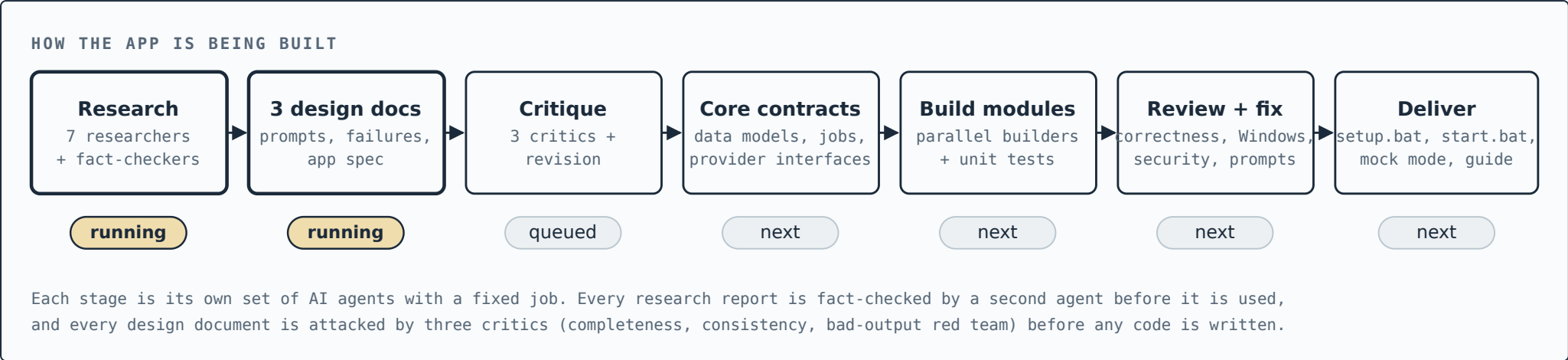
Everything runs on your computer in your browser. Files, history and keys stay on your PC; only AI requests go out, using your own keys. 3D can be made through the Tripo API or by you on Tripo's website: the app always prepares the images and imports the model back.



API KEY	NEEDED?	USED FOR	WHERE TO GET IT
Anthropic (Claude)	Required	plans, critic, yes/no checks, fix instructions (Claude Opus 5 + Sonnet 5)	console.anthropic.com
OpenAI (GPT Image 2.5)	Required	concepts, prints, part views, fabrics. OpenAI may ask you to verify your organization before image models work.	platform.openai.com
Tripo	Required	P2 3D accessories, 4-view images for hair and accessories (paid plan for commercial use)	platform.tripo3d.ai
Recraft	Recommended	clean vector face parts and prints	recraft.ai
Google Gemini	Optional	second opinion when ranking; Nano Banana 2 as a backup image model	aistudio.google.com
fal	Optional	backup models and cloud cut-outs if your PC is slow	fal.ai
Roblox	Not needed	uploads are done by hand in Studio	—

FROM RESEARCH TO A WORKING APP

The same thinking that shapes the skin workflow shapes how the app is built: research first, check every claim, write the plan down, attack it, then build in small tested pieces.



Now Research workflow running: 7 researchers with fact-checkers, then PROMPT_BIBLE.md, FAILURE_MODES.md and APP_SPEC.md, then 3 critics and a revision pass.	Next Core contracts written by hand (data models, job engine, provider interfaces with mock versions), then parallel module builders with unit tests, then a multi-lens review and fixes.	First usable build Settings (keys in Windows Credential Manager), plan loop and Gate 1 concept previews; works in mock mode without keys. Then part board, clothing, Tripo API + manual 3D, checks and export kit.
Five tests to run early Shirt/pants on your own blocky body after upload · head base passes Studio's face check · kit hair passes the validator · image + Tripo pilot day (~\$50) · Studio can take screenshots automatically.	One-time work outside the app Head base (eye shapes, mouths, lids), first 10-15 hairstyles from a 3D artist with full rights, fabric and shading packs, your taste profile (favourites + ~50 ratings).	What stays human About 1-2 hours per duo: approving Gate 1 and Gate 2, hair polish (30-60 min for now), the final pick, and uploading in Roblox Studio (no upload API).